

Software Architecture and Design Assignment

Notes:

- Ensure all group members contribute equally to the assignment.
- Cite all sources appropriately in your written report.
- Report Format: *Softcopy (PDF/Word)*
- Each group must submit their work before the **deadline [April, 22/2026]**.

All groups must address the following four sections for their assigned architecture:

1. Fundamental Concepts (Definitions & Principles)

- Clearly define the core concepts and underlying principles of the architectural style.

2. Design Patterns (Explanation & Examples)

- Explain relevant design patterns (like Singleton, Factory Method, Observer, Strategy) that are commonly used within this architecture.
- Provide clear examples of how these patterns are applied (include code snippets or diagrams where helpful).

3. Architectural Analysis (Pros, Cons, Diagrams)

- Analyze the advantages and disadvantages of this architectural style.
- Discuss the factors that make this architecture a good or bad choice for different scenarios.
- Include at least one clear architectural diagram illustrating the key components and their interactions.

4. Case Study (Real-World Application)

- Identify a well-known software system or company that utilizes this architectural style. Describe how the architecture is implemented in their system.
- Highlight any relevant metrics or outcomes related to their architectural choice.

Software Architecture and Design Assignment

Name	Architecture
1. Surafel Tesfaye 2. Woldemedihnn Mezene 3. Betelhem hailu 4. Kidist wubshet 5. Frezer Getachew	<i>Event-Driven Architecture (EDA)</i> <i>Reactive Architecture</i>
1. Abigia Alemayehu 2. Ermias Tefaye 3. Ruhama Ababu 4. Hasen Ahmed 5. Milkesa Taye 6. Kidus Yonas	<i>Layered Architecture</i> <i>Clean Architecture</i>
1. Natnael esayas 2. Iftu chala 3. Rebira Adugna 4. Yididiya Shimelis 5. Petros Sisay 6. Siham Birhanu	<i>Microkernel Architecture</i> <i>Hexagonal Architecture</i>
1. Lidetu Tesfaye 2. Firomsa Tesfaye 3. Dagmawi Wondwosen 4. Amanuel Getachew 5. Meklit Mengestu 6. Eyuel Mohamued	<i>Space-Based Architecture</i> <i>Peer-To-Peer Architecture</i>
1. Surafel Admas 2. Yididiya Solomon 3. Kenasa Fayera 4. Ibsa Leta 5. Hanan Nuru	<i>Big Data Architecture</i> <i>Pipeline Architecture</i>

Software Architecture and Design Assignment

1. Selamawit Basaznew. 2. Sara Belete. 3. Meaza Yosef. 4. Rahel Tadese. 5. Mihret Tadese. 6. Amanuel Geremu	<i>Multi-Cloud Architecture</i> <i>Security-Centric Architecture</i>
1. Naol gelana 2. Wegari ararssa 3. Ebisa megerssa 4. Abdulletif ylkal 5. Firaol Arrarssa 6. Dagim Girma	<i>Serverless Architecture</i> <i>Microservices Architecture</i>
1. Lidet Admassu 2. Ermias mulugeta 3. Eyuel Getachew 4. Bemnet Awlachew. 5. Frehiwet Zerihun. 6. Abel tagesse	<i>Blackboard Architecture</i> <i>Event-Driven Architecture (EDA)</i>